

Sets And Functions

* Defn 1.1 $\mathbb{N}$: The set of natural numbers

$$\mathbb{N} := \{1, 2, 3, \dots\}$$

* * Well Ordering Property of $\mathbb{N}$: Every subset of $\mathbb{N}$ has a smallest element

* Defn 1.2 A set A is called a proper subset of B , i.e. $A \subset B$, if $A \subset B$ and $A \neq B$.

* Defn 1.3 Set A and B are called disjoint if $A \cap B = \emptyset$

* Defn 1.4 Set theoretic difference of A and B , a.k.a the complement of B relative to A
 $A \setminus B = \{x : x \in A \text{ and } x \notin B\}$

* Lemma 1.5

$$A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$$
$$A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$$

* Theorem 1.6 (Principle of Induction) Let $P(n)$ be a statement that depends upon the natural number n . Suppose that

- (1) $P(1)$ is true
- (2) If $P(n)$ is true then $P(n+1)$ is also true.

Then $P(n)$ is true $\forall n \in \mathbb{N}$

Hint: Proof by contradiction use 1.1

* Defn 1.7 The Cartesian product of two set A and B is defined as

$$A \times B := \{(x, y) : x \in A, y \in B\}$$

* Defn 1.8 A function $f: A \rightarrow B$ is the subset of $A \times B$ s.t for each $x \in A$ $\exists$ a unique $y \in B$ for which $(x, y) \in f$.

$(x, y) \in f$ is written as $f(x) = y$

* Defn 1.9

$$\begin{array}{ccc} \text{Domain} & & \text{Codomain} \\ \text{of } f & & \text{of } f \\ f: \overline{A} & \longrightarrow & \overline{B} \end{array}$$

$$\text{Range of } f \quad R(f) := \{y \in B : \exists x \in A \text{ s.t. } f(x) = y\}$$

Image / Direct Image of $C \subset A$

$$f(C) := \{f(x) : x \in C\}$$

Inverse image of $D \subset B$

$$f^{-1}(D) := \{x \in A : f(x) \in D\}$$

* Lemma 1.10

$f: A \rightarrow B$ and $C, D \subset B$ then

$$(i) \quad f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D)$$

$$(ii) \quad f^{-1}(C \cup D) = f^{-1}(C) \cup f^{-1}(D)$$

$$(iii) \quad f^{-1}(B \setminus C) = A \setminus f^{-1}(C)$$

* Lemma 1.11

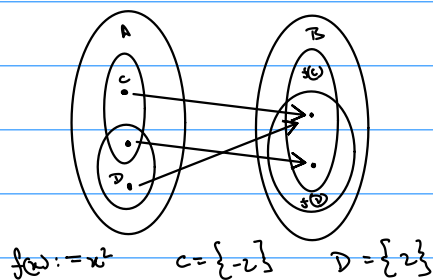
$f: A \rightarrow B$ and $C, D \subset A$ then

$$(i) \quad f(C \cup D) = f(C) \cup f(D)$$

$$(ii) \quad f(C \cap D) \subset f(C) \cap f(D)$$

Proof Hint

Example



* Defn 1.12

A binary relation on a set A is a subset $R \subset A \times A$ which contains the pairs where the relation holds. Instead of $(a, b) \in R$ we write $a R b$.

* Defn 1.13

Sets A and B have the same cardinality if there exists a bijective map $f: A \rightarrow B$.

* Defn 1.14

If a set A has the same cardinality as $\{1, 2, \dots, n\}$ for some $n \in \mathbb{N}$, we say $|A| = n$.

equivalence class of all set with same cardinality as A

} finite cardinality

If A is empty we say $|A| = 0$

*Defn 1.15 $|A| \leq |B|$ $\iff \exists$ a injective map $f: A \rightarrow B$

*Defn 1.16 $\iff |A| = \aleph$, we say A is countably infinite

*Defn 1.17 $\iff A$ is finite or countably infinite we say A is countable, else we say A is uncountable

*Defn 1.18 The power set of set A , denoted by $P(A)$, is the set of all subsets of A .

*Theorem 1.19 $|A| < |P(A)|$, ie there doesn't exist a surjective $f: A \rightarrow P(A)$

Proof by contradiction: Let f be a surjective map

$$S = \{x \in A : x \notin f(x)\}, \text{ let } c \in A \text{ s.t. } f(c) = S$$

Then either $c \in S$ or $c \notin S$

$\mathbb{R}$: Real Numbers

* Defn 2.1 An ordered set S is a set S together with a binary relation $<$ s.t

(i) $\forall x, y \in S$, exactly one is true $<, >$ or $=$

(ii) If $x, y, z \in S$ s.t $x > y$ and $y > z$ then $x > z$

* Defn 2.2 Let $E \subseteq S$ where S is an ordered set

(i) if $\exists b \in S$ s.t $\forall x \in E$ $x \leq b$ then b is called an upper bound.

(ii) if $\exists b_0 \in S$ s.t (a) b_0 is an upper bound

(b) If b is another upper bound then $b_0 \leq b$

then b_0 is called the least upper bound or supremum of E

Similarly the greatest lower bound / infimum of E is defined.

* Defn 2.3 An ordered set S has the least upper bound property if every non empty set of S that is bounded above has a least upper bound.

* Defn 2.4 A set F is called field if it has two operations $(+)$ and $(\cdot)$ defined on it s.t

① If $x, y \in F$ then $x+y \in F$

② $\forall x, y, z \in F$ $(x+y)+z = x+(y+z)$

③ $\exists 0 \in F$ s.t $\forall x \in F$ $x+0 = 0+x = x$

④ $\forall x \in F$ $\exists -x \in F$ s.t $x+(-x) = -x+x = 0$

⑤ $\forall x, y \in F$ $x+y = y+x$

⑥ If $x, y \in F$ then $x \cdot y \in F$

⑦ $\forall x, y, z \in F$ $(x \cdot y) \cdot z = x \cdot (y \cdot z)$

⑧ $\exists 1 \in F$ s.t $\forall x \in F$ $1 \cdot x = x \cdot 1 = x$

⑨ $\forall x \in \{F - \{0\}\}$ $\exists x^{-1} \in F$ s.t $x \cdot x^{-1} = x^{-1} \cdot x = 1$

⑩ $\forall x, y \in F$ $x \cdot y = y \cdot x$

Abelian group
under addition

Abelian group
under multiplication

* Defn 2.5 An ordered field F is an ordered set s.t

(i) If $x, y, z \in F$ s.t $y < z$ then $x+y < x+z$

(ii) If $x > 0$ and $y > 0$ then $xy > 0$

* Lemma 2.6 Let F be an ordered field and $x, y, z, w \in F$ then

- ① If $x > 0$ and $y > z$ then $xy > xz$
- ② If $x \neq 0$ then $x^2 > 0$ Hint $-1 \cdot x = -x$
- ③ $1 > 0$
- ④ $0 < x < y$ then $x^2 < y^2$
- ⑤ $0 < x < y$ then $0 < 1/y < 1/x$

* Lemma 2.7 An ordered field F has the least upper bound property if and only if it has the greatest lower bound property.

* Defn 2.8 Real Number, $\mathbb{R}$, is an unique ordered field with least upper bound property s.t. $\mathbb{Q} \subset \mathbb{R}$.

* Lemma 2.9 If $S \subset \mathbb{R}$ is an upper bounded nonempty set, then $\forall \epsilon > 0 \exists x \in S$ s.t.
 $\sup(S) - \epsilon < x \leq \sup(S)$

* Lemma 2.10 Let $A, B \subset \mathbb{R}$ be nonempty subsets s.t. $x \leq y \forall x \in A$ and $y \in B$, then
 $\sup(A) \leq \inf(B)$

* Theorem 2.11 If $x, y \in \mathbb{R}$ s.t. $x > 0$ and $x < y$ then $\exists n \in \mathbb{N}$ s.t. $nx > y$.
(Archimedian property of $\mathbb{R}$)
Proof by contradiction: Rephrase $\mathbb{N} \subset \mathbb{R}$ is unbounded above
Use 2.9

* Theorem 2.2 $\mathbb{Q}$ is dense in $\mathbb{R}$ i.e. If $x, y \in \mathbb{R}$ s.t. $x < y$ then
 $\exists r \in \mathbb{Q}$ s.t. $x < r < y$
Proof hint Let $x > 0 \exists n \in \mathbb{N}$ s.t. $n(y-x) > 1$. $A := \{k : k \in \mathbb{N} \text{ and } k > nx\}$
 $m \in A$ be the smallest element of A . Then $m > nx \Rightarrow x < m/n$
Also $m-1 < nx \Rightarrow m < nx+1 \Rightarrow m < ny \Rightarrow m/n < y$

*Defn 2.13

Let $A \subset \mathbb{R}$ and $x \in \mathbb{R}$

$$x+A = \{x+y \in \mathbb{R} : y \in A\}$$

$$xA = \{xy \in \mathbb{R} : y \in A\}$$

*Lemma 2.14

If $A \subset \mathbb{R}$ is bounded above, then $\sup(x+A) = x + \sup(A)$

$$\inf(x+A) = x + \inf(A)$$

If $x > 0$, and A is bounded above $\sup(xA) = x \sup(A)$

$$\inf(xA) = x \inf(A)$$

*Defn 2.15

Let $A \subset \mathbb{R}$ be a set

(i) if A is empty then $\sup(A) := -\infty$

$$\inf(A) := \infty$$

(ii) If A is not bounded above then $\sup(A) := \infty$

(iii) if A is not bounded below then $\inf(A) := -\infty$

*Defn 2.16

When $A \subset \mathbb{R}$ is bounded above st $\sup(A) \in A$ then we denote $\sup(A)$ by $\max(A)$. Similarly $\min(A)$ is defined

*Defn 2.17

Absolute value function $f: \mathbb{R} \rightarrow \mathbb{R}^+$

$$|x| := \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

*Lemma 2.18

(i) $|x| \geq 0$ and $|x| = 0$ if and only if $x = 0$

(ii) $|xy| = |x| |y|$

(iii) $|x| \leq y$ if and only if $-y \leq x \leq y$

(iv) $-|x| \leq x \leq |x| \quad \forall x \in \mathbb{R}$

Hint: Divide it into different cases based on signs of x and y

* Lemma 2.19 (Triangle Inequalities)

- (i) $|x+y| \leq |x|+|y| \quad \forall x, y \in \mathbb{R}$
- (ii) $|x-y| \leq |x|+|y| \quad "$
- (iii) $||x|-|y|| \leq |x-y| \quad "$

* Defn 2.20 $f: D \rightarrow \mathbb{R}$. We say f is bounded above if $\exists M \in \mathbb{R}$ s.t.

$$|f(x)| \leq M \quad \forall x \in D$$

* Lemma 2.21 If $f: D \rightarrow \mathbb{R}$ and $g: D \rightarrow \mathbb{R}$ are bounded functions s.t.

$$f(x) \leq g(x) \quad \forall x \in D \text{ then}$$

$$\sup_{x \in D} f(x) \leq \sup_{x \in D} g(x) \quad \inf_{x \in D} f(x) \leq \inf_{x \in D} g(x)$$

* Defn 2.22 Intervals

$$[a, b] := \{ x \in \mathbb{R} \text{ s.t. } a \leq x \leq b \}$$

$$(a, b) := \{ x \in \mathbb{R} \text{ s.t. } a < x < b \}$$

* Lemma 2.23 All intervals have the same cardinality i.e. $\exists$ a bijective mdp b/w any two intervals of $\mathbb{R}$

Example $(-\pi/2, \pi/2) \xrightarrow{\tan(x)} \underbrace{(-\infty, \infty)}_{\mathbb{R}}$

* Theorem 2.24 $\mathbb{R}$ is uncountable

Proof Hint: Let $x \in \mathbb{R}$ be countably infinite s.t. if $a, b \in \mathbb{R}$ and $a < b$ then $\exists x \in X$ s.t. $a < x < b$

$f: \mathbb{N} \rightarrow X$. Let $x_1 := f(1)$ $a_1 := x_1$ $b_1 := x_1 + 1$

Construct $a_1 < a_2 \dots < a_n < b_n < b_{n+1} \dots < b_2 < b_1$ s.t.

$x_j \in (a_k, b_k)$ where $j=1, 2, \dots, k$

$A = \{a_1, a_2, \dots, a_n\}$ $B = \{b_1, b_2, \dots, b_n\}$

Let $y = \sup(A)$ then $y \notin A$, $y \in B$ and $y \neq x$

$\therefore \mathbb{R}$ is uncountable

* Defn 2.25 For an infinite sequence of digits $0.d_1d_2d_3\dots$

$$D_n := \frac{d_1}{10} + \frac{d_2}{10^2} + \dots + \frac{d_n}{10^n}$$

* Lemma 2.26 (i) Every infinite sequence of digits $0.d_1d_2\dots$ represents unique $x \in [0,1]$ st

$$D_n \leq x \leq D_n + \frac{1}{10^n} \quad \forall n \in \mathbb{N}$$

(ii) $\forall x \in (0,1]$ $\exists$ a unique infinite sequence of digits $0.d_1d_2d_3\dots$ that represents x st

$$D_n \leq x \leq D_n + \frac{1}{10^n} \quad \forall n \in \mathbb{N}$$

Sequences And Series

* Defn 3.1 A sequence of real numbers, i.e. $\{x_n\}_{n=1}^{\infty}$ is a function $f: \mathbb{N} \rightarrow \mathbb{R}$

* Defn 3.2 $\{x_n\}_{n=1}^{\infty}$ is called bounded if the underlying function is bounded.
Defn 2.20

* Defn 3.3 $\{x_n\}_{n=1}^{\infty}$ is said to converge to $x \in \mathbb{R}$ if $\forall \epsilon > 0 \exists M \in \mathbb{N}$ s.t.
 $\forall n \geq M \quad |x_n - x| < \epsilon$

Limit of the sequence $\lim_{n \rightarrow \infty} x_n = x$

* Defn 3.4 A sequence that is not convergent is called divergent.

* Lemma 3.5 A convergent sequence has a unique limit.

* Lemma 3.6 A convergent sequence is bounded.

* Lemma 3.7 If $\{x_n\}_{n=1}^{\infty}$ and $\{y_n\}_{n=1}^{\infty}$ are convergent series s.t.
 $x_n \leq y_n \quad \forall n \in \mathbb{N}$ then $\lim_{n \rightarrow \infty} x_n \leq \lim_{n \rightarrow \infty} y_n$

* Defn 3.8 $\{x_n\}_{n=1}^{\infty}$ is called monotone increasing if $x_{n+1} \geq x_n \quad \forall n \in \mathbb{N}$
" " " " decreasing " " $x_{n+1} \leq x_n \quad \forall n \in \mathbb{N}$

* Theorem 3.9 A monotone sequence is bounded if and only if it is convergent.

if $\{x_n\}_{n=1}^{\infty}$ is monotone increasing and bounded $\lim_{n \rightarrow \infty} x_n = \sup \{x_n : n \in \mathbb{N}\}$
" " " " decreasing " " $\lim_{n \rightarrow \infty} x_n = \inf \{x_n : n \in \mathbb{N}\}$

* Lemma 3.10 For $\{x_n\}_{n=1}^{\infty}$ the following statements are equivalent:

(i) $\{x_n\}_{n=1}^{\infty}$ is convergent

(ii) The k tail, i.e. $\{x_{n+k}\}_{n=1}^{\infty}$ converges $\forall k \in \mathbb{N}$

(iii) " " " " " converges for some $k \in \mathbb{N}$

* Defn 3.1 Let $\{n_i\}_{i=1}^{\infty}$ be a strictly increasing sequence of natural numbers.
 $\{x_{n_i}\}_{i=1}^{\infty}$ is called a subsequence of $\{x_n\}_{n=1}^{\infty}$

* Lemma 3.2 If $\{x_n\}_{n=1}^{\infty}$ is a convergent sequence then all its subsequences $\{x_{n_i}\}_{i=1}^{\infty}$ also converge to the same limit

* Squeeze lemma: If $\{x_n\}_{n=1}^{\infty}$, $\{a_n\}_{n=1}^{\infty}$ and $\{b_n\}_{n=1}^{\infty}$ are sequences and $L \in \mathbb{R}$ s.t
 3.13
 (i) $\{a_n\}_{n=1}^{\infty}$ and $\{b_n\}_{n=1}^{\infty}$ are convergent sequences
 (ii) $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} b_n = L$
 (iii) $a_n \leq x_n \leq b_n \quad \forall n \in \mathbb{N}$
 then $\{x_n\}_{n=1}^{\infty}$ also converges to L

* Lemma 3.14 Let $\{x_n\}_{n=1}^{\infty}$, $\{y_n\}_{n=1}^{\infty}$ be convergent sequences then
 (i) $\{z_n\}_{n=1}^{\infty}$ where $z_n = x_n + y_n$ converges to $\lim_{n \rightarrow \infty} z_n = \lim_{n \rightarrow \infty} x_n + \lim_{n \rightarrow \infty} y_n$
 (ii) $\{z_n\}_{n=1}^{\infty}$ where $z_n = x_n y_n$ converges to $\lim_{n \rightarrow \infty} z_n = \lim_{n \rightarrow \infty} x_n \lim_{n \rightarrow \infty} y_n$

Proof Hint: Let $\epsilon > 0$. $K = \max(|x_1|, |y_1|, 1)$ $\exists M_1 \in \mathbb{N}$ s.t $\forall n > M_1 \quad |x_n - x| < \epsilon/3K$
 $\exists M_2 \in \mathbb{N}$ s.t $\forall n > M_2 \quad |y_n - y| < \epsilon/3K$

$$\forall n > \max(M_1, M_2) \quad \text{Prove} \quad |x_n y_n - xy| = |(x_n - x)(y_n - y) + x(y_n - y) + (x_n - x)y| < \epsilon$$

(ii) if $y_n \neq 0 \quad \forall n \in \mathbb{N}$ and $\lim_{n \rightarrow \infty} y_n \neq 0$ then $\{z_n\}_{n=1}^{\infty}$ where $z_n = x_n / y_n$ converges to $\lim_{n \rightarrow \infty} z_n = \lim_{n \rightarrow \infty} x_n / \lim_{n \rightarrow \infty} y_n$

Proof Hint: Prove $\{1/y_n\}_{n=1}^{\infty}$ converges to $1/y$ where $y = \lim_{n \rightarrow \infty} y_n$

$$\text{Let } \epsilon > 0 \quad \exists M_1 \in \mathbb{N} \text{ s.t } |y_n - y| < \min\left(|y|^2 \frac{\epsilon}{2}, \frac{|y|}{2}\right)$$

$$\forall n > M_1 \quad |y| = |y - y_n + y_n| \leq |y - y_n| + |y_n| < \frac{|y|}{2} + |y_n| \Rightarrow \frac{1}{|y_n|} < \frac{2}{|y|}$$

$$\left| \frac{1}{y_n} - \frac{1}{y} \right| = \left| \frac{y - y_n}{y y_n} \right| = \frac{|y - y_n|}{|y| |y_n|} < \frac{|y - y_n| \epsilon/2}{|y|} \times \frac{2}{|y|} < \epsilon$$

* Lemma 3.15 If $\{x_n\}_{n=1}^{\infty}$ is a convergent sequence s.t. $x_n \geq 0 \quad \forall n \in \mathbb{N}$, then

$$(i) \quad \lim_{n \rightarrow \infty} \sqrt{x_n} = \sqrt{\lim_{n \rightarrow \infty} x_n}$$

$$(ii) \quad \lim_{n \rightarrow \infty} |x_n| = \left| \lim_{n \rightarrow \infty} x_n \right|$$

* Lemma 3.16 Let $\{x_n\}_{n=1}^{\infty}$ be a sequence, $x \in \mathbb{R}$ and $\{a_n\}_{n=1}^{\infty}$ be another sequence s.t.

(i) $\{a_n\}_{n=1}^{\infty}$ is convergent

$$(ii) \quad \lim_{n \rightarrow \infty} a_n = 0$$

$$(iii) \quad |x_n - x| < a_n \quad \forall n \in \mathbb{N}$$

then $\{x_n\}_{n=1}^{\infty}$ is convergent and $\lim_{n \rightarrow \infty} x_n = x$

* Lemma 3.17 Let $c > 0$

(i) If $c < 1$, the sequence $\{c^n\}_{n=1}^{\infty}$ converges to 0

(ii) If $c > 1$, " " " is unbounded

* Lemma 3.18 Let $\{x_n\}_{n=1}^{\infty}$ be a sequence s.t.

$$(i) \quad x_n \neq 0 \quad \forall n \in \mathbb{N}$$

$$(ii) \quad L = \lim_{n \rightarrow \infty} \frac{|x_{n+1}|}{|x_n|} \text{ exists}$$

then (a) if $L < 1$ then the sequence $\{x_n\}_{n=1}^{\infty}$ converges to 0

(b) " $L > 1$ " " " is unbounded

* Defn 3.19 Let $\{x_n\}_{n=1}^{\infty}$ be a bounded sequence. Define $\{a_n\}_{n=1}^{\infty}$ and $\{b_n\}_{n=1}^{\infty}$ sequences where

$$a_n := \sup \{x_k : k \geq n\} \quad b_n := \inf \{x_k : k \geq n\}$$

$$\text{Limit superior} \quad \limsup_{n \rightarrow \infty} x_n := \lim_{n \rightarrow \infty} a_n$$

$$\text{Limit inferior} \quad \liminf_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} b_n$$

Lemma 3.20

For a bounded sequence $\{x_n\}_{n=1}^{\infty}$

$$\limsup_{n \rightarrow \infty} x_n := \lim_{n \rightarrow \infty} a_n = \inf \{a_n : n \in \mathbb{N}\}$$

$$\liminf_{n \rightarrow \infty} x_n := \lim_{n \rightarrow \infty} b_n = \sup \{b_n : n \in \mathbb{N}\}$$

Also $\liminf_{n \rightarrow \infty} x_n \leq \limsup_{n \rightarrow \infty} x_n$

Theorem 3.21

If $\{x_n\}_{n=1}^{\infty}$ is a bounded sequence, then $\exists$ subsequences $\{x_{n_i}\}_{i=1}^{\infty}, \{x_{m_j}\}_{j=1}^{\infty}$ s.t.

$$\limsup_{n \rightarrow \infty} x_n = \lim_{i \rightarrow \infty} x_{n_i}$$

Need not to be monotone

$$\liminf_{n \rightarrow \infty} x_n = \lim_{j \rightarrow \infty} x_{m_j}$$

Proof Hint:

Let $n_1 = 1$ Let $n_1, n_2, \dots, n_{k-1}$ are already defined

$m \in \mathbb{N}$ s.t. m s.t. $\forall n \geq m \quad |a_{n_{k-1}+1} - x_n| < 1/k$ Define $n_k = m$

then $\forall k \geq 2 \quad |a_{n_k} - x_{n_k}| < \frac{1}{k}$ as $a_{n_k} \leq a_{n_{k-1}+1}$

$\{a_{n_k}\}_{k=1}^{\infty}$ is subsequence of $\{a_n\} \rightarrow \lim_{n \rightarrow \infty} a_n = \lim_{k \rightarrow \infty} a_{n_k} = a$

Let $\epsilon > 0$, $\exists M_1 \in \mathbb{N}$ s.t. $\forall i \geq M_1, |a_{n_i} - a| < \epsilon/2$

$$M \in \mathbb{N} \text{ s.t. } 1/M_2 < \epsilon/2 \quad j = \max(M_1, M_2, 2)$$

$$\forall i \geq j \quad |x_{n_i} - a| \leq |x_{n_i} - a_{n_i}| + |a_{n_i} - a| < \frac{1}{i} + \epsilon/2 < \frac{1}{j} + \epsilon/2 \leq \frac{1}{M_2} + \epsilon/2 < \epsilon/2 + \epsilon/2 = \epsilon$$

Theorem 3.22

Let $\{x_n\}_{n=1}^{\infty}$ be a bounded sequence. Then $\{x_n\}_{n=1}^{\infty}$ converges if and only if

$$\liminf_{n \rightarrow \infty} x_n = \limsup_{n \rightarrow \infty} x_n$$

Further $\liminf_{n \rightarrow \infty} x_n = \limsup_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} x_n$

Proof Hint: Use 3.21, 3.11
Use 3.12

Lemma 3.23

If $\{x_n\}_{n=1}^{\infty}$ is a bounded sequence and $\{x_{n_k}\}_{k=1}^{\infty}$ is a subsequence, then

$$\liminf_{n \rightarrow \infty} x_n \leq \liminf_{k \rightarrow \infty} x_{n_k} \leq \limsup_{k \rightarrow \infty} x_{n_k} \leq \limsup_{n \rightarrow \infty} x_n$$

* Lemma 3.24 A bounded sequence $\{x_n\}_{n=1}^{\infty}$ is convergent to x if and only if every convergent subsequence $\{x_{n_k}\}_{k=1}^{\infty}$ converges to x
Use 3.11, 3.21 and 3.13

* Theorem 3.25 Bolzano-Weierstrass: Suppose a sequence $\{x_n\}_{n=1}^{\infty}$ of real numbers is bounded. Then there exists a convergent subsequence $\{x_{n_k}\}_{k=1}^{\infty}$

* Defn 3.26 $\{x_n\}_{n=1}^{\infty}$ diverges to infinity, i.e. $\lim_{n \rightarrow \infty} x_n = \infty$ if $\forall K \in \mathbb{R} \exists M \in \mathbb{N}$ s.t.
 $\forall n \geq M \quad x_n > K$

$\{x_n\}_{n=1}^{\infty}$ diverges to $-\infty$, i.e. $\lim_{n \rightarrow \infty} x_n = -\infty$ if $\forall K \in \mathbb{R} \exists M \in \mathbb{N}$ s.t.
 $\forall n \geq M \quad x_n < K$

* Defn 3.27 Extension of defn of lim sup and lim inf to unbounded sequences

* Defn 3.28 A sequence $\{x_n\}_{n=1}^{\infty}$ is called a Cauchy sequence. If $\forall \epsilon > 0, \exists m \in \mathbb{N}$ s.t.
 $\forall k, m \geq m \quad |x_m - x_k| < \epsilon$

* Lemma 3.29 If a sequence is Cauchy, it is bounded

* Theorem 3.30 A sequence of real numbers is Cauchy if and only if it is convergent.

Using 3.21
Proof Hint Cauchy $\Rightarrow$ bounded $\Rightarrow \exists \{x_{n_i}\}_{i=1}^{\infty}$ s.t. $\lim_{i \rightarrow \infty} x_{n_i} = \limsup_{n \rightarrow \infty} x_n = a$
 $\hookrightarrow \exists \{x_{m_j}\}_{j=1}^{\infty}$ s.t. $\lim_{j \rightarrow \infty} x_{m_j} = \liminf_{n \rightarrow \infty} x_n = b$

$$\begin{aligned} \exists M_1 \in \mathbb{N} \text{ s.t. } \forall i \geq M_1 \quad |x_{n_i} - a| &< \epsilon/3 & |a - b| &= |a - x_{n_i} + x_{n_i} - x_{m_i} + x_{m_i} - b| \\ \exists M_2 \in \mathbb{N} \text{ s.t. } \forall i \geq M_2 \quad |x_{m_i} - b| &< \epsilon/3 & &\leq |x_{n_i} - a| + |x_{n_i} - b| + |x_{n_i} - x_{m_i}| \\ \exists M_3 \in \mathbb{N} \text{ s.t. } \forall i \geq M_3 \quad |x_{n_i} - x_{m_i}| &< \epsilon/3 & &< \epsilon/3 + \epsilon/3 + \epsilon/3 < \epsilon \end{aligned}$$

Let $i = \max(M_1, M_2, M_3)$

$\therefore a = b$
Use 3.13

* Defn 3.31 Given a sequence $\{x_n\}_{n=1}^{\infty}$, a series is defined as
$$\sum_{n=1}^{\infty} x_n.$$

A series is said to converge if the sequence of partial sums $\{s_k\}$ converges where
$$s_k := \sum_{n=1}^k x_n = x_1 + x_2 + \dots + x_k$$

* Lemma 3.32 (K-tail) Let $\sum_{n=1}^{\infty} x_n$ be a series and $M \in \mathbb{N}$, then $\sum_{n=1}^{\infty} x_n$ converges if and only if $\sum_{n=M}^{\infty} x_n$ converges

* Defn 3.33 A series $\sum_{n=1}^{\infty} x_n$ is called Cauchy if the sequence of partial sums $\{s_k\}$ is Cauchy.

* Lemma 3.34 The series $\sum_{n=1}^{\infty} x_n$ is Cauchy if and only if $\forall \epsilon > 0 \exists$ an $M \in \mathbb{N}$ s.t $\forall n \geq M$ and every $k > n$
$$\left| \sum_{i=n+1}^k x_i \right| < \epsilon$$

* Lemma 3.35 Let $\sum_{n=1}^{\infty} x_n$ be a convergent series then the sequence $\{x_n\}_{n=1}^{\infty}$ is convergent to 0, i.e
$$\lim_{n \rightarrow \infty} x_n = 0$$

* Lemma 3.36 Let $\kappa \in \mathbb{R}$ and $\sum_{n=1}^{\infty} x_n, \sum_{n=1}^{\infty} y_n$ be convergent series then

(i) $\sum_{n=1}^{\infty} \kappa x_n$ is also a convergent series and further

$$\sum_{n=1}^{\infty} \kappa x_n = \kappa \sum_{n=1}^{\infty} x_n$$

(ii) $\sum_{n=1}^{\infty} (x_n + y_n)$ is a convergent series and further

$$\sum_{n=1}^{\infty} (x_n + y_n) = \sum_{n=1}^{\infty} x_n + \sum_{n=1}^{\infty} y_n$$

* Defn 3.37 Absolute convergence: A series $\sum_{n=1}^{\infty} x_n$ is said to converge absolutely if the series $\sum_{n=1}^{\infty} |x_n|$ converges.

A series is called conditionally convergent if it is convergent but not absolutely convergent.

* Lemma 3.38 If a series $\sum_{n=1}^{\infty} x_n$ is absolutely convergent then it is also convergent

* Lemma 3.39 Let $\sum_{n=1}^{\infty} x_n$ and $\sum_{n=1}^{\infty} y_n$ are series s.t. $0 \leq x_n \leq y_n \quad \forall n \in \mathbb{N}$ then
(comparison test) (i) If $\sum_{n=1}^{\infty} y_n$ is convergent then $\sum_{n=1}^{\infty} x_n$ is also convergent
Use 3.9, 3.6

(ii) If $\sum_{n=1}^{\infty} x_n$ diverges then $\sum_{n=1}^{\infty} y_n$ also diverges.

* Lemma 3.40 The series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ where $p \in \mathbb{R}$ is convergent if and only if $p > 1$

* Lemma 3.41 Let $\sum_{n=1}^{\infty} x_n$ is a series s.t. $x_n \neq 0 \quad \forall n \in \mathbb{N}$ and $L = \lim_{n \rightarrow \infty} \frac{|x_{n+1}|}{|x_n|}$ then
(Ratio test)

(i) If $L < 1$ then $\sum_{n=1}^{\infty} x_n$ converges absolutely Read Hint + Use 3.38

(ii) If $L > 1$ then $\sum_{n=1}^{\infty} x_n$ diverges

Continuous Functions

* Defn 4.1

Let $S \subset \mathbb{R}$. Then $x \in \mathbb{R}$ is called a cluster point of S if $\forall \varepsilon > 0$

the $S \cap (x - \varepsilon, x + \varepsilon) \setminus \{x\}$ is non empty

Rephrasing $\forall \varepsilon > 0 \exists y \in S$ s.t. $y \neq x$ and $|x - y| < \varepsilon$

* Lemma 4.2

Let $S \subset \mathbb{R}$ and then $x \in \mathbb{R}$ is a cluster point of S if and only if there exists

$\{x_n\}_{n=1}^{\infty}$ s.t. $\forall n \ x_n \neq x, x_n \in S$ and $\lim_{n \rightarrow \infty} x_n = x$.

Proof hint use 3.16 with $a_n := x_n$

* Defn 4.3

Let $S \subset \mathbb{R}$, $c \in \mathbb{R}$ be a cluster point of S , $f: S \rightarrow \mathbb{R}$, $L \in \mathbb{R}$.

We say $f(x) \rightarrow L$ as $x \rightarrow c$ or $\lim_{x \rightarrow c} f(x) = L$

if $\forall \varepsilon > 0, \exists \delta > 0$ s.t.

if $x \in S \setminus \{c\}$ and $|x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon$

* Lemma 4.4

The limit of a function $f(x)$ as $x \rightarrow c$ if it exists is unique

* Lemma 4.5

Let $S \subset \mathbb{R}$ and c be its cluster point. Let $f: S \rightarrow \mathbb{R}$ be a function.

(Sequental Limit)

$f(x) \rightarrow L$ as $x \rightarrow c$ if and only if for every sequence $\{x_n\}_{n=1}^{\infty}$ s.t. $\forall n \in \mathbb{N} \ x_n \in S \setminus \{c\}$ and $\lim_{n \rightarrow \infty} x_n = c$, the sequence $\{f(x_n)\}_{n=1}^{\infty}$ converges to L .

Proof hint:

$\underbrace{x \rightarrow c \quad f(x) \rightarrow L}_A$

$\underbrace{\text{Every } \{x_n\}_{n=1}^{\infty} \text{ s.t. } \forall n \ x_n \in S \setminus \{c\} \lim_{n \rightarrow \infty} x_n = c \text{ then } \lim_{n \rightarrow \infty} f(x_n) = L}_B$

step 1 If A is true B is true

step 2 If A is false B is false

$\exists$ an $\varepsilon > 0$ s.t. $\forall \delta > 0 \exists x \in S \setminus \{c\}$ s.t. $|x - c| < \delta$ and $|f(x) - L| \geq \varepsilon$

Construct $\{x_n\}_{n=1}^{\infty}$

Let $\delta = \gamma_n$

$\underbrace{|x_n - c| < \gamma_n}_{\lim_{n \rightarrow \infty} x_n = c}$

$\underbrace{|f(x_n) - L| \geq \varepsilon}_{\{f(x_n)\}_{n=1}^{\infty} \text{ does not converge to } L}$

* Corollary
4.6

Let $S \subset \mathbb{R}$, c be a cluster point of S . Suppose $f: S \rightarrow \mathbb{R}$ and $g: S \rightarrow \mathbb{R}$ be functions s.t. the limits of $f(x)$ and $g(x)$ exists as $x \rightarrow c$, and $f(x) \leq g(x) \quad \forall x \in S$
then $\lim_{x \rightarrow c} f(x) \leq \lim_{x \rightarrow c} g(x)$

* Corollary
4.7

Let $S \subset \mathbb{R}$ and c be a cluster point of S . Suppose $f: S \rightarrow \mathbb{R}$, $g: S \rightarrow \mathbb{R}$, $h: S \rightarrow \mathbb{R}$ are functions s.t. $g(x) \leq f(x) \leq h(x) \quad \forall x \in S$.

Suppose $\lim_{x \rightarrow c} g(x)$ and $\lim_{x \rightarrow c} h(x)$ exist and $\lim_{x \rightarrow c} g(x) = \lim_{x \rightarrow c} h(x)$

then $\lim_{x \rightarrow c} f(x)$ exist and $\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} g(x) = \lim_{x \rightarrow c} h(x)$

Corollary
4.8

Let $S \subset \mathbb{R}$, c be a cluster point of S . Suppose $f: S \rightarrow \mathbb{R}$, $g: S \rightarrow \mathbb{R}$ be functions and $\lim_{x \rightarrow c} f(x)$ and $\lim_{x \rightarrow c} g(x)$ exists. Then

$$(i) \quad \lim_{x \rightarrow c} (f(x) + g(x)) = \left(\lim_{x \rightarrow c} f(x) \right) + \left(\lim_{x \rightarrow c} g(x) \right)$$

$$(ii) \quad \lim_{x \rightarrow c} (f(x) \cdot g(x)) = \left(\lim_{x \rightarrow c} f(x) \right) \left(\lim_{x \rightarrow c} g(x) \right)$$

$$(iii) \quad g(x) \neq 0 \quad \forall x \in S / \{c\} \quad \text{and} \quad \lim_{x \rightarrow c} g(x) \neq 0$$

$$\lim_{x \rightarrow c} \left(\frac{f(x)}{g(x)} \right) = \frac{\lim_{x \rightarrow c} f(x)}{\lim_{x \rightarrow c} g(x)}$$

Corollary
4.9

Let $S \subset \mathbb{R}$, c be a cluster point of S . Suppose $f: S \rightarrow \mathbb{R}$ is a function s.t. as $x \rightarrow c$ the limit of $f(x)$ exists. Then

$$\lim_{x \rightarrow c} |f(x)| = \left| \lim_{x \rightarrow c} f(x) \right|$$

* Defn 4.10 Let $f: S \rightarrow \mathbb{R}$ and $A \subset S$. Define the function $f|_A: A \rightarrow \mathbb{R}$ by
 (Restriction of f to A)

$$f|_A(x) := f(x) \quad \text{for } x \in A$$

* Lemma 4.11 Let $S \subset \mathbb{R}$, $c \in \mathbb{R}$ and $f: S \rightarrow \mathbb{R}$. Suppose $A \subset S$ s.t. $\exists \alpha > 0$ for which

$$(A \setminus \{c\}) \cap (c-\alpha, c+\alpha) = (S \setminus \{c\}) \cap (c-\alpha, c+\alpha)$$

(i) c is a cluster point of A if and only if it is a cluster point of S

(ii) Suppose c is a cluster point of S then $f(x) \rightarrow L$ as $x \rightarrow c$ if and only if

$$f|_A(x) \rightarrow L \text{ as } x \rightarrow c$$

Proof hint if $A \cap C = B \cap C$ then if

$$\supset C \text{ then } A \cap D = B \cap D$$

* Defn 4.12 Let $f: S \rightarrow \mathbb{R}$, $c \in \mathbb{R}$ be a cluster point of $S \cap (c, \infty)$ and the limit of the restriction of f to $S \cap (c, \infty)$ exists then

$$\lim_{x \rightarrow c^+} f(x) := \lim_{x \rightarrow c} f|_{S \cap (c, \infty)}(x)$$

* Lemma 4.13 Let $S \subset \mathbb{R}$, c is a cluster point of both $S \cap (c, \infty)$ and $S \cap (-\infty, c)$, $f: S \rightarrow \mathbb{R}$, $L \in \mathbb{R}$. Then (i) c is a cluster point of S

$$(ii) \lim_{x \rightarrow c} f(x) = L \quad \text{if and only if} \quad \lim_{x \rightarrow c^+} f(x) = \lim_{x \rightarrow c^-} f(x) = L$$

* Defn 4.14 Let $S \subset \mathbb{R}$, $c \in S$ and $f: S \rightarrow \mathbb{R}$. f is called continuous at c if $\forall \epsilon > 0 \exists \delta > 0$ s.t. if $x \in S$ and $|x - c| < \delta$ then $|f(x) - f(c)| < \epsilon$

* Defn 4.15 $f: S \rightarrow \mathbb{R}$ is called continuous if it is continuous $\forall c \in S$

* Lemma 4.16 Let $S \subset \mathbb{R}$, $f: S \rightarrow \mathbb{R}$, $c \in S$. then

(i) if c is not a cluster point of S then f is continuous at c .

(ii) if c is a cluster point of S , then f is continuous at c if and only if $\lim_{x \rightarrow c} f(x) = f(c)$

(iii) f is continuous at c if and only for all $\{x_n\}_{n=1}^{\infty}$ where $x_n \in S$ and $\lim_{n \rightarrow \infty} x_n = c$ the sequence

$$\{f(x_n)\}_{n=1}^{\infty} \text{ converges to } f(c)$$

* Lemma 4.17

Let $S: S \rightarrow \mathbb{R}$ and $g: S \rightarrow \mathbb{R}$ be continuous functions at $c \in S$. then

(i) $h: S \rightarrow \mathbb{R}$ $h(x) := f(x) + g(x)$ is continuous at c

(ii) " " " $h(x) := f(x) - g(x)$ " " " "

(iii) If $g(x) \neq 0 \quad \forall x \in S$ then $h: S \rightarrow \mathbb{R}$ $h(x) := \frac{f(x)}{g(x)}$ is continuous at c .

* Lemma 4.18

Let $A, B \subset \mathbb{R}$, $f: B \rightarrow \mathbb{R}$, $g: A \rightarrow B$. If g is continuous at $c \in A$ and f is continuous at $g(c)$ then $f \circ g: A \rightarrow \mathbb{R}$ $f \circ g(x) := f(g(x))$ is continuous at c .

* Lemma 4.19

A continuous function $f: [a, b] \rightarrow \mathbb{R}$ is bounded

* Theorem 4.20
Extreme value theorem

A continuous function $f: [a, b] \rightarrow \mathbb{R}$ on a closed and bounded interval $[a, b]$ achieves both an absolute maximum and an absolute minimum on $[a, b]$

* Lemma 4.21

Let $f: [a, b] \rightarrow \mathbb{R}$ be a continuous function. Suppose $f(a) < 0$ and $f(b) > 0$, then $\exists c \in (a, b)$ s.t. $f(c) = 0$

* Intermediate Value theorem 4.22

Let $f: [a, b] \rightarrow \mathbb{R}$ be a continuous function. Suppose $y \in \mathbb{R}$ is s.t. $f(a) < y < f(b)$ or $f(a) > y > f(b)$. Then $\exists c \in (a, b)$ s.t. $f(c) = y$

* Corollary 4.23

If $f: [a, b] \rightarrow \mathbb{R}$ is a continuous function then the direct image $f([a, b])$ is a closed and bounded interval or a single number

